



Differential Learning as a Key Training Approach to Improve Creative and Tactical Behavior in Soccer

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ABSTRACT

Purpose: The aim of this study was to identify the effects of a differential-learning program, embedded in small-sided games, on the creative and tactical behavior of youth soccer players. Forty players from under-13 (U13) and under-15 (U15) were allocated into control and experimental groups and were tested using a randomized pretest to posttest design using small-sided games situations. **Method:** The experimental group participated in a 5-month differential-learning program embodied in small-sided games situations, while the control group participated in a typical small-sided games training program. In-game creativity was assessed through notational analyses of the creative components, and the players' positional data were used to compute tactical-derived variables. **Results:** The findings suggested that differential learning facilitated the development of creative components, mainly concerning attempts (U13, small; U15, small), versatility (U13, moderate; U15, small), and originality (U13, unclear; U15, small) of players' actions. Likewise, the differential-learning approach provided a decrease in fails during the game in both experimental groups (moderate). Moreover, differential learning seemed to favor regularity in pitch-positioning behavior for the distance between players' dyads (U13, small; U15, small), the distance to the team target (U13, moderate; U15, small), and the distance to the opponent target (U13, moderate; U15, small). **Conclusions:** The differential-learning program stressed creative and positional behavior in both age groups with a distinct magnitude of effects, with the U13 players demonstrating higher improvements over the U15 players. Overall, these findings confirmed that the technical variability promoted by differential learning nurtures regularity of positioning behavior.

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Creative behavior is a higher-order disposition in team sports, and several studies have confirmed that creativity is trainable (Memmert, 2015). More recently, the creativity developmental framework has been proposed as a suitable reference model that provides general guidelines to boost creative behavior throughout the early years (Santos, Memmert, Sampaio, & Leite, 2016). Creativity refers to players' disposition of movement and attunement outside the box under the guidance of the environment and their ability to solve specific game problems in a novel, feasible, unexpected, and original way by starting a single act or flowing in a collective action that will lead to their team's success (Santos, Memmert, et al., 2016). Accordingly, coaches should be aware that during the sport preparation process, creative behavior assumes different manifestations characterized as P-creativity (personal) and H-creativity (historical). During the early years, P-creativity prevails because this expression is internal to the player and is related to the exploration of new behaviors to overcome individual limitations. Otherwise, H-creativity is a behavior

widely recognized as an innovation that no player has ever executed before and is commonly connected to a high level of expertise (Hristovski, Davids, Araujo, & Passos, 2011). Therefore, the exploratory behavior has a determinant role to nurture these creative expressions (Hristovski et al., 2011; Hristovski, Davids, Passos, & Araujo, 2012; Santos, Memmert, et al., 2016) and embraces players' efforts to perform unlimited movement configurations under specific variations, rather than execute a predetermined solution (Hristovski et al., 2012).

A sports environment that encourages exploratory behavior will probably lead to an increase in the emergence of creative components, such as attempts, fluency, versatility, and originality. Attempts are recognized as any effort to perform different actions even if noneffective. Fluency or efficacy is the ability to execute as many effective movement actions as possible. Versatility or flexibility is the ability to produce different actions, whereas originality is the ability to generate new and unique actions that others are not likely to perform (for reference, see Santos,

Memmert, et al., 2016). Measuring creativity in team sports is challenging, and commonly, previous components are placed in evidence to assess creative behavior in team sports. For instance, Santos and colleagues (2017) found positive effects of a creativity sports-based training program (Skills4genius), which was sustained in diversification, physical literacy, teaching games for understanding, and differential learning on creative behavior development. Overall, their study results confirmed that the enrichment program enhanced fluency, attempts, and versatile actions during the game in children without previous experiences in structured sports. These results reinforce the importance of using training approaches that provide freedom to explore and adapt to the properties of an environment (Chow, Davids, Button, & Renshaw, 2015).

Based on the previous considerations, differential learning seems to emerge as a promising approach to enhance creative behavior (Memmert, 2015; Santos et al., 2017). Developed on the dynamical systems perspective, differential learning has been mostly applied in the context of motor learning (Frank, Michelbrink, Beckmann, & Schöllhorn, 2008; Schöllhorn et al., 2006; Schöllhorn, Michelbrink, Welmski, & Davids, 2009). This approach is characterized by increasing the number of movement fluctuations through no movement repetition and without corrections during the skill acquisition process (Schöllhorn, Hegen, & Davids, 2012). Therefore, the main idea is to amplify the noise using movement variability to stress the limits of performing novel movement solutions (Schöllhorn et al., 2006). The role of random variability in allowing players to acquire new and functional movement patterns could be necessary to unleash creativity. Actually, the differential-learning approach might be especially relevant for youth players who should be exposed early to suitable opportunities to acquire functional and transferable movement skills (Santos, Mateus, Sampaio, & Leite, 2016). Within the differential-learning scope, players are usually instructed to actively perform movement errors, instead of avoiding them, to promote constant adaptations (e.g., kick a ball with arms raised; Schöllhorn et al., 2006). Therefore, considering the impossibility of performing two identical movements during a match, the utilization of traditional methods of learning, characterized by endless repetitions toward an ideal movement and sustained in error corrections, should be reconsidered (Schöllhorn et al., 2009).

In this vein, the differential-learning approach has been related to higher skill acquisition and better learning rates over traditional methods in many sports (Henz & Schöllhorn, 2016; Schöllhorn et al., 2009; Wagner & Muller, 2008). In soccer, these effects have been confirmed in goal-shooting and ball-control interventions

(Schöllhorn et al., 2006, 2012). However, available research in this area is still scarce and might be appropriate to assess differential-learning effects in dynamic and ecological contexts that require constant adaptation, such as small-sided games (SSG). In fact, SSG concurrently foster technical and tactical behaviors (Hill-Haas, Dawson, Impellizzeri, & Coutts, 2011) as well as exploratory behavior (Torrents et al., 2016). Recent research focused on soccer players' tactical behavior has received considerable attention (Barnabe, Volossovitch, Duarte, Ferreira, & Davids, 2016; Gonçalves et al., 2017a; Memmert, Lemmink, & Sampaio, 2017). Several studies have analyzed how players adapt their behavior according to specific modifications in SSG (Folgado, Duarte, Fernandes, & Sampaio, 2014) and how different interpersonal relations patterns that emerge during performance are functionally different (Folgado, Duarte, et al., 2014; Gonçalves et al., 2017b). For example, a study by Sampaio and Maças (2012) revealed the effects of a soccer training program on players' positional behavior. The results showed that players' behavior became more regular after a 13-week training program, possibly as a consequence from an increase in game-tactical principles and pitch-positioning awareness. To find this evidence, the approximate entropy (ApEn) computation technique has been commonly used to quantify the structure of the variability, which allows researchers to identify if players' positioning expresses a regular pattern (Harbourne & Stergiou, 2009). Nevertheless, further research seems to be needed to understand how movement variations provided by the differential-learning approach applied in game-based situations impacts the technical and positional-related variables. Therefore, the aim of this study was to identify the effects of a differential-learning program in the creative and tactical behavior of youth players during soccer SSG. It was expected that the variability promoted by the differential-learning approach nurtures the components of creativity that are more related to the performance of different and novel actions and that it favors the regularity of tactical behavior.

Method

Participants

Participants were 40 Portuguese youth soccer players from under-13 (U13) and under-15 (U15) teams with at least 2 years of experience in structured sports (including soccer or other types of sport). The players belonged to two different teams competing at a regional level. During the sports season, all players performed three training sessions per week (90–115 min) and played an official game during the weekend with durations of 60 min and

70 min for the U13 and U15 teams, respectively. Participants were randomly allocated into two experimental groups of U13 players ($n = 10$; $M_{\text{age}} = 11.1 \pm 0.5$ years; $M_{\text{height}} = 150.5 \pm 7.8$ cm; $M_{\text{weight}} = 43.1 \pm 5.2$ kg; $M = 4.4 \pm 2.9$ years of experience in soccer; $M = 1.4 \pm 0.5$ sports practiced, including soccer) and U15 players ($n = 10$; $M_{\text{age}} = 13.1 \pm 0.3$ years; $M_{\text{height}} = 161.3 \pm 8.3$ cm; $M_{\text{weight}} = 54.4 \pm 4.7$ kg; $M = 7.1 \pm 1.5$ years in soccer; $M = 1.2 \pm 0.4$ sports practiced) who participated in the differential-learning training program and two corresponding control groups of U13 players ($n = 10$; $M_{\text{age}} = 11.4 \pm 0.5$ years; $M_{\text{height}} = 154.0 \pm 6.1$ cm; $M_{\text{weight}} = 32.9 \pm 13.3$ kg; $M = 5.3 \pm 2.5$ years in soccer; $M = 1.7 \pm 0.9$ sports practiced) and U15 players ($n = 10$; $M_{\text{age}} = 13.0 \pm 0.8$ years; $M_{\text{height}} = 161.4 \pm 7.9$ cm; $M_{\text{weight}} = 57.2 \pm 9.1$ kg; $M = 6.8 \pm 1.6$ years in soccer; $M = 1.5 \pm 0.7$ sports practiced). Goalkeepers (Gk) were part of the study but were excluded from the analysis to avoid misleading interpretations once their positioning was restricted. Parents, coaches, and the club were fully informed of the aims and procedures of the study and signed informed consent forms for children's participation. All participants were notified that they could withdraw from the study at any time. The study protocol was approved by and followed the guidelines stated by the local Institutional Research Ethics Committee and conformed to the recommendations of the Declaration of Helsinki.

Procedure

All participants took part in a familiarization session 1 week before the testing sessions. Afterward, each group performed two testing sessions, one for the pretest and

another for the posttest. Before each testing session, there was a standardized 15-min warm-up based on running and ball possession games without goals (5 v 5). Individual and collective game behavior were assessed during a Gk + 5-v-5 + Gk soccer SSG played on an artificial-turf pitch measuring 36 m $\times$ 25 m (length $\times$ width) using 7-v-7 goals. The SSG protocol was composed of two bouts of 6 min interspersed with 3 min of passive recovery. The head coach divided the players into two balanced teams, and the players played each SSG according to their usual playing position role. All SSG were performed as much as possible with the official game rules, except the offside rule, which was not applied. Several balls were placed around the field to ensure replacement as fast as possible. Coach intervention was minimized, and therefore, no feedback was allowed during the game.

Training intervention

The experimental groups participated in a differential-learning training program for 5 months (40 sessions) with a volume of ~ 90 min per week. The training program was composed of 3 training sessions per week, each with a duration of 30 min and performed at the beginning of participants' teams' training sessions. Afterward, participants joined the rest of their team to continue their soccer training schedule (for the next ~ 60 minutes). Assumptions between the typical SSG performed by the control teams and the differential-learning SSG followed by the experimental teams are outlined in Figure 1. Also, the movement variations used during the differential-learning training program are presented in Table 1, and some were combined (i.e., shot while consecutively throwing a tennis ball with one eye covered). Different numerical relations were

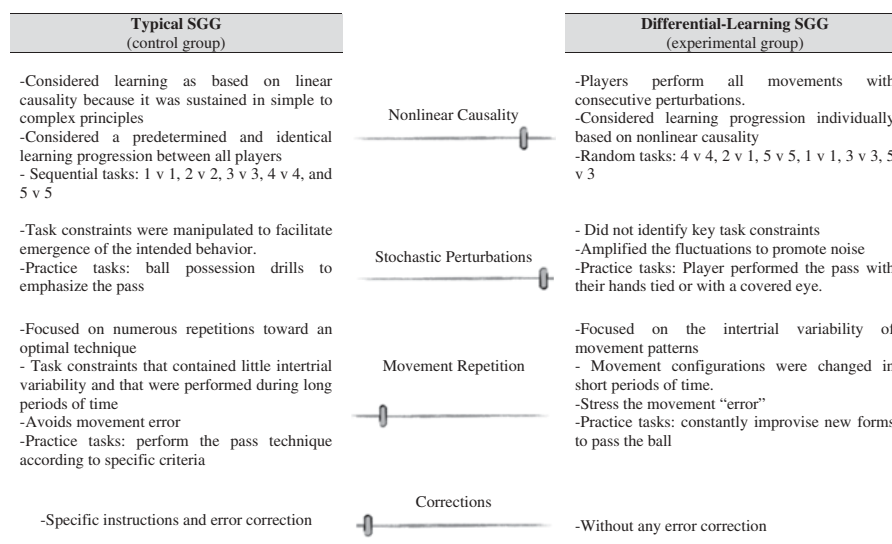


Figure 1. Methodological differences between typical and differential-learning small-sided games (SSG).

Table 1. Example of the variations performed in traditional and differential-learning small-sided games (SSG).

Game-based variables		Typical SSG (control group)	Differential-learning SSG (experimental group)
Player's Number	Numerical Relations	Balanced (1 v 1, 2 v 2, 3 v 3)	Balanced and unbalanced (4 v 3, 2 v 1, 3 v 3, 4 v 5, 5 v 5)
Pitch	Progressions	Sequential (from 1 v 1 to 5 v 5)	Random variations
	Playing surface	Artificial turf	Artificial turf and natural grass
	Shape	Rectangle and square	Rectangle, square, triangle, diamond, and circle
Playing Ball	Size	Small, medium, large	Small, medium, large
	Type	Normal size	Smaller, normal-size, and larger-size.
	Number of balls	Soccer ball	Soccer, tennis, handball, rugby, reflex ball, futsal, and fitball.
Targets	Number	Just one	One and two simultaneously
	Size	1 to 4	1 to 4
	Type of Scoring	Small targets and 7 v 7	Small targets (pop-up), 7-v-7 and 11-v-11 targets
Field Obstacles	Disposition	Stop ball, shot	Stop ball, shot, pick the ball into a box
		Final line of the pitch	In the final line, inside and out of the pitch
Body Constraints	Upper part of body	None	Sticks, ropes, hoop, cones, tchoukball targets, and barriers
	Lower part of body	None	Visual occlusion, hands behind head, both arms up, hands tied, rotating arms forward, hands on hips, play with 3D glasses, give touches with a tennis ball on the hand, juggling a balloon, elastic bands, bracelets, players connected with an elastic band
Games Rules		Limited touches	Play with nondominant limb, with ankles, without trainers, in each ball contact improvise different types of technical skills
		Add neutral players	Double points if different passes, dribbles, or shoots were performed
		Full versus low pressure	Divide the pitch into spaces and limit the specific space allowed to play.
Combinations		None	During the game, two or more types of the previous variations were combined.

used within the SSG and ranged from 1 v 1 to 5 v 5. However, different schedules were designed according to the control or experimental groups (for more details, see Table 1). For example, while in the control groups SSG were played using balanced teams in a sequential order, during the differential-learning program, there were random scenarios, such as numerical inferiority, as it seems to facilitate exploratory behavior (Torrents et al., 2016). Considering that the program was sustained by different SSG situations, the number of repetitions for pass, dribble, and shot conducted per participant varied between sessions.

Additionally, to characterize the remaining session, all groups followed typical soccer training sessions three times a week, with each session lasting ~ 90 minutes. The coaches and the club provided only a general description of the training guidelines for each age group. Moreover, the following description is only related to the tasks in which players were engaged after the differential-learning or typical SSG training programs. In this sense, after the training program, the U13 training sessions were composed of 40 min of game principles and tactical organization and finished with 20 min of SSG or continuous play (7 v 7). The U15 training sessions were focused on the development of fundamental tactical skills. Thus, after the training program, they performed tactical training tasks with a duration of 30 min, and then the session finished with 30 min of SSG or continuous play (11 v 11). An experienced coach, two assistant coaches, and a

researcher fully supervised the training sessions. The study started in the middle of the competitive season and lasted until the end of the respective season.

Measures

The Creative Behavior Assessment in Team Sports

The SSG were recorded using two digital cameras from a fixed position located in the superior plane (3 m above) to cover the entire pitch and positioned 45° from one of the goal lines. The digital cameras' frequency was 25 Hz, and resolution was 1,280 pixels × 720 pixels. Individual creative behavior was assessed through a computerized notational analysis using LongoMatch software (Version 1.3.2, Fluendo, Barcelona, Spain). Afterward, the data were organized in a pre-prepared spreadsheet (Excel for Windows®) entitled Creative Behavior Assessment in Team Sports (CBATS). The CBATS was developed to measure creativity in ball possession during game performance (Santos et al., 2017). Measurements included pass, dribble, and shot skills, which were divided into successful and unsuccessful actions (see Liu, Gomez, Goncalves, & Sampaio, 2016). Thus, fails and attempts were considered the unsuccessful actions, while fluency, versatility, and originality comprised the successful actions. A successful pass referred to an intentional played ball from one player to a teammate player, while an unsuccessful pass was when the pass does not reach the teammate (i.e., ball

Table 2. Description of the creative components observed through the Creative Behavior Assessment in Team Sports.

Creative Components			Behavioral Descriptions
Pass	Unsuccessful	Fail Attempts	A standardized pass does not reach the intended teammate The pass does not reach the receiver, but the player tries to explore a nonstandardized action (noneffective)
		Successful	Fluency
	Successful	Versatility	The pass reaches the receiver but in a nonstandardized way (effective and different)
		Originality	The player considers the rarity of overall versatile passes performed during the game
		Standardized behavior	The player performs a pass with the dominant limb and uses the inside part of the foot
		Nonstandardized behavior	The player considers all executions that are different from the standardized pass (i.e., pass with head)
Dribble	Unsuccessful	Fails	Player in possession of the ball fails to overcome the defender when realizing a standardized dribble
		Attempts	Player fails to progress in the pitch or gain a favorable position when realizing a nonstandardized dribble
	Successful	Fluency	Player dribbles the opponent toward progress in the pitch or moves toward a more favorable position using a standardized dribble
		Versatility	Player dribbles the opponent toward progress in the pitch or moves toward a more favorable position in a nonstandardized way
		Originality	Player considers the unusualness of versatile dribbles performed during the game
	Standardized behavior	Player dribbles around the opponent using a simple side touch with the dominant limb	
	Nonstandardized behavior	Player considers all executions that are different from the standardized dribble (i.e., roulette)	
	Shot	Unsuccessful	Fails
Attempts			A nonstandardized off-target shot
Successful		Fluency	Standardized on-target shot (the finalization is effective)
		Versatility	Nonstandardized on-target shot (the finalization is effective and different)
		Originality	Player considers the rarity of versatile shots performed during the game
Standardized behavior		Off-target and on-target shot with dominant limb, in which the player hatted the ball with the instep, top or inside of the foot.	
Nonstandardized behavior	Player considers all executions or variations that were different from the standardized shot		
Absolute Values (frequency)			Measuring the creative components was considered the frequency of the actions

interception by the opponent team). A successful dribble was defined as an initiative by the player with the ball to explore a duel against his direct opponent or opponents to gain space to perform a pass or score while retaining possession of the ball. This action was regarded as unsuccessful when the player in possession of the ball could not overcome the defending player (i.e., dribbler is tackled). Regarding the shot, successful actions (shot on target) were considered an effort to score a goal with any regular part of the body, in which the ball hit the target or required Gk intervention. In contrast, the shot off target was considered an unsuccessful action.

Previous variables were classified as embodied in the creativity components (see Table 2), such as attempts, fluency, versatility, and originality, which have been used to describe creative behavior in team sports (Santos, Memmert, et al., 2016). In this study, originality took into account the criteria related to the rareness of the versatile actions; for example, the versatile actions that players only performed once regarding the pretest and posttest game situations for the U13 and U15 teams were considered to be original. Additionally, this study added

fails measurements to understand if a training program sustained in differential learning decreases unsuccessful actions. Analysis of the previous five components during SSG formats allowed us to capture qualitative changes as the degrees of freedom of retained movements' performance, which is not possible when movement is observed as a single technique in isolation, usually applied during the differential-learning protocols. Moreover, all components were assessed considering absolute values of occurrence or frequency. An experienced sport performance analyst gathered the data, and reliability was inspected by retesting 17.5% of the sample 1 week later and assessing fails, attempts, fluency, versatility, and originality of pass, dribble, and shot skills. The obtained intrarater reliability coefficient was high and considered reliable (intra-class correlation > .84; O'Donoghue, 2010).

Collective behavior (positional performance)

Positional data during SSG were gathered using 5-Hz global positioning system units (SPI-PRO, GPSports, Canberra, Australian Capital Territory, Australia). The units were placed into appropriate elastic harnesses on each participant's upper back. Geodetic coordinates were

exported from the units and computed using dedicated routines in Matlab® (MathWorks, Natick, MA) according to previous data corrections guidelines (Folgado, Duarte, et al., 2014). Players' dynamic positional data were used to compute the distance between dyads (i.e., the interpersonal distance between each pair of teammates Gonçalves et al., 2017b), the distance to the team's own target (i.e., the distance from each player to their own target, using the middle two-dimensional [2D] point coordinates of the target as reference), and the distance to the opponent team target (i.e., the distance from each player to the middle 2D point coordinates of the opponent 2D target; Vilar et al., 2014). All previous variables were calculated by computing the norm between the vectors using the following equation (Equation 1),

$$D(a_{x(t),y(t)}, b_{x(t),y(t)}) = \sqrt{(a_{x(t)} - b_{x(t)})^2 + (a_{y(t)} - b_{y(t)})^2} \quad (1)$$

where D is the distance, a is the player, x and y are the coordinates, t is the time, and b is the player's teammate, own target, or their opponent's target. Moreover, distances were also analyzed according to amount of variability expressed by the coefficient of variation and according to the structure of the variability using ApEn (Gonçalves et al., 2017b). From the procedure perspective, the imputed values used to compute the ApEn were 2 to vector length (m) and 0.2 standard deviations to tolerance (r ; Yentes et al., 2013). The values from ApEn range from 0 to 2, with the lower values representing more repeatable patterns (e.g., a value near 0 in the distance to the team's own target, which means that the players tended to repeat the distance to the target in a repeatable fashion). Thus, ApEn might be useful to understand the regularity and predictability of movement behavior, which may, in turn, provide information regarding the functionality of dynamic players' positioning (Gonçalves et al., 2017b).

Data analysis

After preliminary inspections for outliers, distribution, and assumptions, data were processed to fit a 2×2 factorial analysis of covariance (ANCOVA; age: U13, U15; group: control, experimental), with posttest values as the dependent variable and pretest values as the covariate. The statistical analysis was performed using the Statistical Package for the Social Sciences software (SPSS Inc., Chicago, IL), and statistical significance was set as $p < .05$. In addition, all pretest and posttest values were compared between teams via standardized Cohen

differences, computed with pooled variance and respective 95% confidence limits. Thresholds for effect size statistics were 0.2 = trivial, 0.6 = small, 1.2 = moderate, 2.0 = large, and > 2.0 , very large (Hopkins, Marshall, Batterham, & Hanin, 2009).

Results

The summary of findings from the ANCOVA for age, group, and age/group factors are presented in Table 3. In addition, the overall results related to the effects of the differential-learning training program in-game creative components of U13 and U15 age groups are presented in Figures 2a and 2b and Table 4, respectively. Regarding the fails measurements, the ANCOVA results revealed significant main effects between groups, $F(4, 65) = 5.5$, $p < .05$. Hence, a decrease from the pretest to the posttest in both the U13 and U15 experimental teams was identified (standardized Cohen differences $\pm$ 95% confidence limits = -1.10 ± 0.81 and -0.82 ± 0.54 , both moderate). In the experimental U13 group, the fails decrease was more pronounced in the dribble (-0.61 ± 0.76 , moderate) and in the shot (-0.82 ± 0.73 , moderate), while in the experimental U15 group, it was more pronounced in the pass (-0.55 ± 0.55 , moderate) and shot actions (-0.48 ± 0.61 , small). Interestingly, both the U13 and U15 experimental teams presented a small increase in their attempts (0.58 ± 0.72 and 0.22 ± 0.40 , respectively), whereas a moderate increase was identified only in the U13 control team. Accordingly, attempts were mostly stressed during pass actions in all the previous groups (U13 control, 0.51 ± 0.49 ; U13 experimental, 0.58 ± 0.72 ; U15 experimental, 0.58 ± 0.56 , small).

In addition, the differential-learning training program did not provide any benefits in fluency because all the results were unclear, except in the U15 control group, who showed a small decrease. On the other hand, versatility could be considered the variable most stressed by the training program with a clearly statistical difference, $F(4, 65) = 15.3$, $p < .01$ between groups. This difference was due to the unclear results revealed by both the U13 and U15 control teams, which is in contrast to the increase displayed by both experimental teams (U13, 0.88 ± 0.63 , moderate; and U15, 0.49 ± 0.60 , small). The increase in versatility in the U13 group was distributed over all the actions (pass, 0.60 ± 0.54 , moderate; dribble, 0.89 ± 0.54 , moderate; and shot, 0.62 ± 0.72 , moderate); in turn, the U15 group only improved in versatile passes (0.90 ± 0.63 , moderate). Nevertheless, the dribble was the variable that displayed more statistical differences according to age and group factors (see Table 3). These results are

Table 3. Main effects for age, group, and the interaction of Age $\times$ Group for the study variables (analysis of covariance results).

Variables		df	Age		Group		Age $\times$ Group	
			F	p	F	p	F	p
Creative Component s	Fails	(4, 65)	0.02	.87	5.47	.02	1.38	.24
	Attempts	(4, 65)	3.84	.05	0.08	.76	3.76	.05
	Fluency	(4, 65)	0.01	.92	1.92	.17	0.80	.37
	Versatility	(4, 65)	1.53	.22	15.27	< .01	0.09	.75
	Originality	(4, 65)	1.61	.20	5.90	< .01	0.13	.71
Pass	Fails_pass	(4, 65)	3.95	.05	0.06	.79	0.22	.63
	Attempts_pass	(4, 65)	2.67	.10	1.81	.18	2.43	.12
	Fluency_pass	(4, 65)	3.25	.07	1.90	.17	0.37	.54
	Versatility_pass	(4, 65)	0.77	.38	16.41	< .01	1.02	.31
	Originality_pass	(4, 65)	0.01	.90	4.40	.04	0.01	.90
Dribble	Fails_dribble	(4, 65)	6.21	< .01	21.32	< .01	6.07	< .01
	Attempts_dribble	(4, 65)	1.52	.22	0.35	.55	1.52	.22
	Fluency_dribble	(4, 65)	1.27	.26	0.98	.32	0.00	.93
	Versatility_dribble	(4, 65)	8.83	< .01	7.41	< .01	3.72	.05
	Originality_dribble	(4, 65)	3.94	.05	2.35	.13	1.02	.31
Shot	Fails_shot	(4, 65)	0.23	.63	0.81	.36	0.02	.88
	Attempts_shot	(4, 65)	0.73	.39	0.00	.94	1.19	.27
	Fluency_shot	(4, 65)	6.41	< .01	0.01	.89	0.02	.86
	Versatility_shot	(4, 65)	0.68	.41	2.29	.13	0.00	.96
	Originality_shot	(4, 65)	0.49	.48	0.31	.57	0.31	.57
Positional	Dyads (m)	(4, 155)	0.66	.41	8.74	< .01	1.69	.19
	Dyads (CV)	(4, 155)	0.05	.82	7.45	< .01	0.04	.82
	Dyads (ApEn)	(4, 155)	0.17	.67	0.52	.47	6.16	< .01
	Target (m)	(4, 75)	1.30	.25	0.17	.67	0.00	.97
	Target (ApEn)	(4, 75)	11.71	< .01	4.08	.04	1.63	.20
	Opp Target (m)	(4, 75)	0.16	.68	0.70	.40	0.24	.62
	Opp Target (ApEn)	(4, 75)	12.44	< .01	5.34	.02	1.40	.24

Note. Values in bold represent significant differences at $p < .05$. m = meters; CV = coefficient of variation; ApEn = approximate entropy; df = degrees of freedom; Opp = opponent.

due to the fact that the dribble was the action most stressed by the U13 experimental group, while no effects were identified in the U15 experimental group and no trend was identified in both control groups. Finally, original actions revealed statistical differences between groups, $F(4, 65) = 5.9$, $p < .01$, because the U15 experimental team showed a small increase (0.51 ± 0.49). Originality in the U15 group was mainly performed in pass actions (0.46 ± 0.53 , small).

The effects of the differential training program in positional variables are presented in Figures 3a and 3b. Statistical differences in group comparisons were identified in the distance between dyads, $F(4, 155) = 8.7$, $p < .01$. While there was a decrease in the distance between dyads for the U15 experimental group (-0.22 ± 0.25 , small), the opposite was found in the respective control group (0.34 ± 0.18 , small). This tendency was more evident regarding the coefficient of variation in the distance between dyads, $F(4, 155) = 7.4$, $p < .01$, whereas both control groups decreased their variation and both experimental groups showed an increase. Nevertheless, the ApEn of the distance between dyads revealed differences according to age/group, $F(4, 155) = 6.1$, $p < .01$, because both control groups displayed opposite trends. Instead, the differences in the team and the opponent's target were emphasized while considering age, $F(4, 75) = 11.7/12.4$, $p < .01$, respectively, and group, $F(4, 75) = 4.1/3.4$, $p < .05$, respectively. In fact, these results could be explained, whereas the U13 experimental team showed a decrease in the

ApEn values of the distance between dyads, team, and the opponent target (U13 experimental, -0.55 ± 0.37 , small; -1.02 ± 0.53 , moderate; -1.12 ± 0.56 , moderate, respectively; U13 control, -0.31 ± 0.35 , small; -0.04 ± 0.75 , unclear, 0.02 ± 0.72 unclear, respectively). The same trend was found for the U15 experimental groups but with a minor magnitude in all previous variables (U15 experimental, -0.54 ± 0.39 , small; -0.43 ± 0.58 , small; -0.36 ± 0.55 , small, respectively, and U15 control, -0.68 ± 0.34 , moderate; -0.11 ± 0.43 , unclear; -0.15 ± 0.40 , unclear, respectively).

Discussion

Current literature has shown several findings related to the benefits of differential learning on skill acquisition in team sports. This first report expands these findings by providing results that identified the differential-learning effects in game creativity and tactical behavior during SSG situations in youth soccer players. Overall, the results support the assumption that a differential-learning training program facilitates the development of creativity components, such as attempts, versatility, and originality of movement actions. Likewise, there was a substantial decrease in fails during the game performance, and it seems that the increase in variability promoted by the differential-learning approach favored the regularity of positional behavior compared with the typical SSG formats.

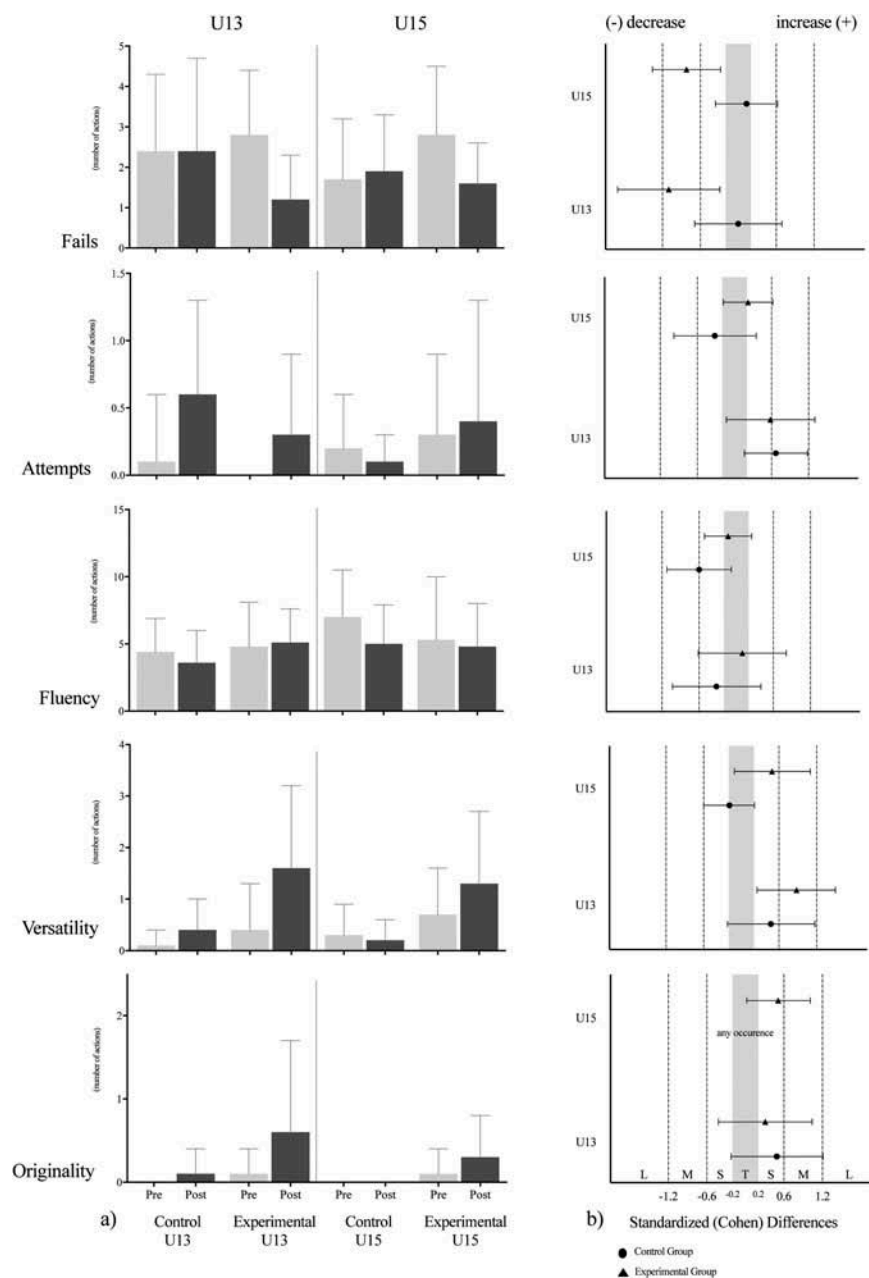


Figure 2. (a) Descriptive (mean $\pm$ SD) and (b) inferential (Cohen's $d \pm$ 95% confidence limit) analysis for attempts, fluency, versatility, and originality components between control and experimental groups on under-13 (U13) and under-15 (U15) teams. *Note.* T = trivial; S = small; M = moderate; L = large.

Analysis of the results confirmed that differential learning demonstrated higher effectiveness in the U13 and U15 players' predisposition to perform versatile actions during a game. Additionally, it has also increased the players' ability to explore unsuccessful but nonstandardized actions (attempts), and these results seem to be in line with several studies suggesting that differential learning is more beneficial because it stresses that players explore a variety of novel movement configurations (Frank et al., 2008; Henz & Schöllhorn, 2016; Schöllhorn et al., 2006, 2012; Wagner & Muller, 2008). In fact, the learning process

is not merely a passive copying process, but instead, it is sustained in the players' exploration, which favors P-creativity (Hristovski et al., 2011, 2012). Moreover, the functional variability and adaptation demands provided by the differential-learning assumptions support the emergence of creative components and encourage players to unleash their creativity (Memmert, 2015; Santos et al., 2017; Santos, Memmert, et al., 2016). The previous increase in creative components might be a consequence of the pronounced intertrial variability provided by differential learning (Schöllhorn et al., 2012). By continuously confronting youth players with

Table 4. Descriptive (mean $\pm$ SD) and inferential (Cohen's $d \pm$ 95% confidence limit) analysis for the creative components, divided into pass, dribble, and shot actions, with comparisons between the control and experimental groups in the under-13 (U13) and under-15 (U15) teams.

Variables		Control group		Cohen's $d \pm$ 95% confidence limit	Experimental group		Cohen's $d \pm$ 95% confidence limit
		Pretest (mean $\pm$ SD)	Posttest (mean $\pm$ SD)		Pretest (mean $\pm$ SD)	Posttest (mean $\pm$ SD)	
U13	Fails	1.1 $\pm$ 1.3	0.6 $\pm$ 0.7	-0.52 $\pm$ 0.72	1.4 $\pm$ 1.4	0.8 $\pm$ 0.9	-0.56 $\pm$ 0.77
Pass	Attempts	0.1 $\pm$ 0.3	0.3 $\pm$ 0.6	0.51 $\pm$ 0.49	0.0 $\pm$ 0.0	0.3 $\pm$ 0.6	0.58 $\pm$ 0.72
	Fluency	2.6 $\pm$ 1.8	1.7 $\pm$ 1.0	-0.61 $\pm$ 0.59	4.1 $\pm$ 3.1	3.3 $\pm$ 2.6	-0.29 $\pm$ 0.65
	Versatility	0.1 $\pm$ 0.3	0.1 $\pm$ 0.3	0.0 $\pm$ 0.74	0.2 $\pm$ 0.4	0.6 $\pm$ 0.9	0.60 $\pm$ 0.54
	Originality	0.0 $\pm$ 0.0	1.0 $\pm$ 0.0	—	0.1 $\pm$ 0.3	0.3 $\pm$ 0.8	0.31 $\pm$ 0.73
U13	Fails	1.0 $\pm$ 1.4	1.6 $\pm$ 1.5	0.41 $\pm$ 0.64	0.8 $\pm$ 1.0	0.3 $\pm$ 0.5	-0.61 $\pm$ 0.76
Dribble	Attempts	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	—	0.0 $\pm$ 0.0	0.1 $\pm$ 0.3	0.34 $\pm$ 0.72
	Fluency	0.8 $\pm$ 1.0	0.8 $\pm$ 1.2	0.05 $\pm$ 0.77	0.3 $\pm$ 0.6	0.5 $\pm$ 0.7	0.36 $\pm$ 0.52
	Versatility	0.1 $\pm$ 0.3	0.3 $\pm$ 0.5	0.49 $\pm$ 0.76	0.3 $\pm$ 0.6	1.0 $\pm$ 1.0	0.89 $\pm$ 0.54
	Originality	0.0 $\pm$ 0.0	0.1 $\pm$ 0.3	0.34 $\pm$ 0.72	0.1 $\pm$ 0.3	0.3 $\pm$ 0.5	0.81 $\pm$ 0.76
U13	Fails	0.3 $\pm$ 0.7	0.3 $\pm$ 0.5	-0.10 $\pm$ 0.73	0.6 $\pm$ 0.6	0.1 $\pm$ 0.3	-0.82 $\pm$ 0.73
Shot	Attempts	0.1 $\pm$ 0.3	0.1 $\pm$ 0.3	-0.00 $\pm$ 0.74	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	—
	Fluency	1.0 $\pm$ 1.2	1.1 $\pm$ 0.9	0.05 $\pm$ 0.69	0.4 $\pm$ 0.7	0.9 $\pm$ 1.1	0.50 $\pm$ 0.55
	Versatility	0.0 $\pm$ 0.0	0.1 $\pm$ 0.3	0.34 $\pm$ 0.72	0.0 $\pm$ 0.0	0.2 $\pm$ 0.4	0.62 $\pm$ 0.72
	Originality	0.0 $\pm$ 0.0	0.1 $\pm$ 0.3	0.34 $\pm$ 0.72	0.0 $\pm$ 0.0	0.1 $\pm$ 0.3	0.41 $\pm$ 0.90
U15	Fails	1.0 $\pm$ 1.1	1.1 $\pm$ 1.0	0.14 $\pm$ 0.59	1.8 $\pm$ 1.3	1.1 $\pm$ 1.0	-0.55 $\pm$ 0.55
Pass	Attempts	0.1 $\pm$ 0.3	0.0 $\pm$ 0.0	-0.44 $\pm$ 0.64	0.1 $\pm$ 0.2	0.3 $\pm$ 0.5	0.58 $\pm$ 0.56
	Fluency	5.4 $\pm$ 2.5	4.0 $\pm$ 2.2	-0.58 $\pm$ 0.40	4.4 $\pm$ 4.2	4.0 $\pm$ 3.0	-0.12 $\pm$ 0.38
	Versatility	0.2 $\pm$ 0.4	0.1 $\pm$ 0.3	-0.14 $\pm$ 0.30	0.2 $\pm$ 0.4	0.9 $\pm$ 1.0	0.90 $\pm$ 0.63
	Originality	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	—	0.1 $\pm$ 0.2	0.2 $\pm$ 0.4	0.46 $\pm$ 0.53
U15	Fails	0.4 $\pm$ 0.6	0.5 $\pm$ 0.8	0.14 $\pm$ 0.52	0.3 $\pm$ 0.8	0.2 $\pm$ 0.4	-0.17 $\pm$ 0.69
Dribble	Attempts	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	—	0.2 $\pm$ 0.6	0.1 $\pm$ 0.3	-0.38 $\pm$ 0.43
	Fluency	1.2 $\pm$ 1.1	0.6 $\pm$ 1.0	-0.49 $\pm$ 0.52	0.3 $\pm$ 0.5	0.3 $\pm$ 0.6	0.0 $\pm$ 0.54
	Versatility	0.2 $\pm$ 0.4	0.1 $\pm$ 0.3	-0.14 $\pm$ 0.52	0.4 $\pm$ 0.6	0.3 $\pm$ 0.7	-0.25 $\pm$ 0.77
	Originality	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	—	0.1 $\pm$ 0.2	0.1 $\pm$ 0.2	0.0 $\pm$ 0.0
	Fails	0.4 $\pm$ 0.6	0.3 $\pm$ 0.8	-0.07 $\pm$ 0.57	0.7 $\pm$ 1.1	0.2 $\pm$ 0.6	-0.48 $\pm$ 0.61
U15	Attempts	0.1 $\pm$ 0.2	0.1 $\pm$ 0.2	0.0 $\pm$ 0.65	0.0 $\pm$ 0.0	0.1 $\pm$ 0.3	0.46 $\pm$ 0.67
Shot	Fluency	0.6 $\pm$ 0.8	0.5 $\pm$ 0.8	-0.18 $\pm$ 0.72	0.7 $\pm$ 1.0	0.6 $\pm$ 0.7	-0.26 $\pm$ 0.56
	Versatility	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	—	0.1 $\pm$ 0.2	0.1 $\pm$ 0.5	0.14 $\pm$ 0.69
	Originality	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	—	0.0 $\pm$ 0.0	0.1 $\pm$ 0.2	0.32 $\pm$ 0.67

different movement variations, motor system adaptation is kept constantly active and unlocks a whole range of behaviors that are potentially available to the players (Frank et al., 2008; Schöllhorn et al., 2006). In fact, the functional role of adaptive movement variability seems necessary to improve performance in youth soccer players (Chow et al., 2015; Hristovski et al., 2011, 2012; Seifert, Button, & Davids, 2013).

Interestingly, the training program seemed to be more beneficial for the U13 team because this age group demonstrated higher rates of improvement over the U15 team in the creative components. Commonly, players in early stages of learning naturally display greater exploratory search behavior. These findings are in line with the available literature, which supports a decrease in creativity thinking with age, possibly as a result of the increased focus on convergent thinking tasks (Kim, 2011; Santos, Memmert, et al., 2016). A similar interpretation could be established in sports settings. As age increases, there is a progressive focus on tactical training arrangements and team organization with a clear decrease in the exploration of individual actions (Folgado, Lemmink, Frencken, & Sampaio, 2014). As a consequence, this decrease in movement variability may impair players' ability to adapt to the

requirements of the environment and possibly affects long-term development preparation (Davids, Glazier, Araujo, & Bartlett, 2003; Santos, Mateus, et al., 2016; Seifert et al., 2013). Indeed, previous reports have shown that talented players were not only dependent on when to perform a skill, but also how to perform it (Kannekens, Elferink-Gemser, & Visscher, 2011). Therefore, this evidence demonstrated that coaches can use differential-learning approaches during their training sessions to foster players' adaptive behavior to unpredictable situations. Moreover, they can give opportunities to youth players to challenge their boundaries, possibly making them more adaptive without being afraid to take risks in competitive situations (Santos et al., 2017; Santos, Memmert, et al., 2016).

Nonetheless, the differential-learning training program promoted different extents in the pass, dribble, and shot between the U13 and U15 age groups. For instance, there was more distributed improvement in all technical skills among the U13 players. Perspectives from long-term development models have suggested the age of 13 years is the trainability boundary to learn new movement behaviors in sport (Stafford, 2005). In addition, it has also been previously reported that the differential-learning approach leads to flexible

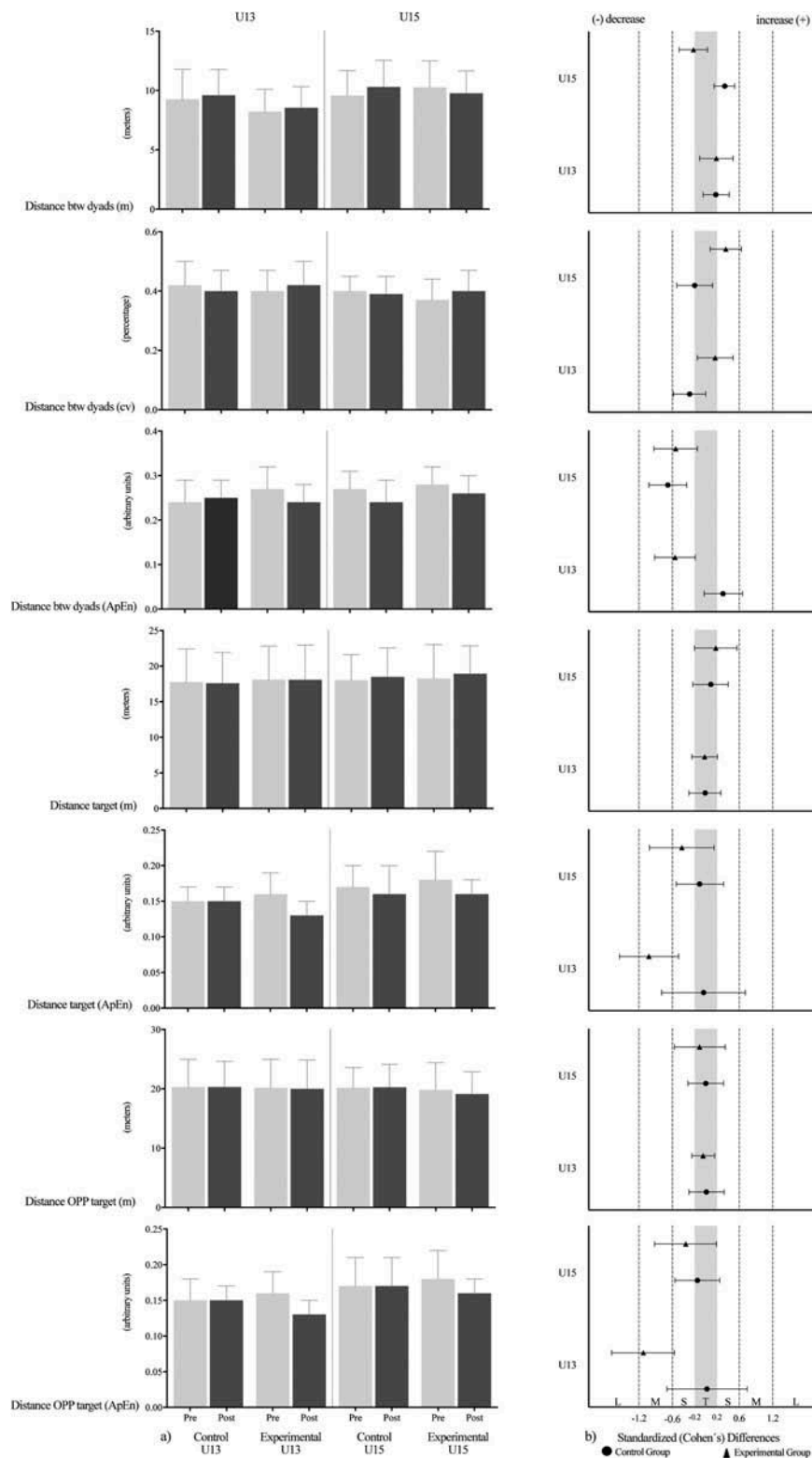


Figure 3. (a) Descriptive (mean $\pm$ SD) and (b) inferential (Cohen's $d \pm$ 95% confidence limit) analysis for positional variables between control and experimental groups on under-13 (U13) and under-15 (U15) teams. *Note.* ApEn = approximate entropy; OPP = opponent; T = trivial; S = small; M = moderate; L = large.

movement solutions (Santos et al., 2017). Therefore, this evidence may help to explain the increase in new passing attempts and in versatile passes, dribbles, and

shots. In contrast, among the U15 players, the pass was the skill most stressed by the differential-learning training program. In this group, players' ability to attempt

new forms of pass and to explore versatile and original passes was an evident marker from the differential-learning training program. These results may be linked to commitment to the game principles and players' expertise. Accordingly, evidence has shown that players of a younger age are likely to adopt behaviors that allow them to quickly approach opponents' targets, such as dribbles or shots. Otherwise, older players are likely to show a more rational positional disposition and develop more organized ball possession toward opponents' goals using the pass as the main action (Barnabe et al., 2016; Folgado, Lemmink, et al., 2014). Therefore, it is not surprising that among the U15 players, the pass emerged as the preferable functional movement behavior to achieve the task goal. These findings indicate that the differential-learning training program had a tendency to stress the most frequent skills used in the SSG formats, which vary according to players' expertise levels.

On the other hand, the impact of differential learning was not equally great on all creative components. As expected, the differential-learning program proved to be less sensitive in fluency development for both age groups. These results were predictable, as the core idea of the differential-learning approach is to boost movement variability instead of fostering standardized actions (Frank et al., 2008; Schöllhorn et al., 2006, 2012). In spite of the lack of fluency observed in differential-learning groups at least, no decrease was found, as it was observed in the U15 typical SSG group. Therefore, differential learning not only provided functional variability in movement configurations, but it also maintained the level of fluency. An additional advantage of differential learning was the greater decrease in fail actions. In the U13 age group, this decrease was more pronounced in the dribble and shot, while in the U15 group, it was more pronounced in the pass and shot. As opposed to the purpose of SSG with a more typical learning method, differential learning seems to induce self-organized learning in which players improve performance according to their individual limitations. Further, youth soccer players are confronted with a higher number of movement variations and without general corrections, the likelihood of them accomplishing their needs is considerably higher (Schöllhorn et al., 2012).

In team sports, the game context dynamically generates decision demands that players should attempt to produce functional solutions. Thus, the learning process should also be grounded in inducing tactical problems within a wide adaptive environment to foster creative behavior (Memmert, 2015; Santos et al., 2017). With regards to the outcomes of the positional

variables, a clear tendency was observed using the ApEn technique. This processing technique allows for measuring the probability that a behavior pattern has to be repeated over time and consequently becomes more regular. In this study, the values from the ApEn technique highlighted that players presented more regular positional behaviors after the differential-learning training program in the U13 and U15 teams comparatively with typical SSG. In general, these intentional positional adjustments may reveal a better understanding of the game, whereas the players showed higher regularity in their behaviors according to the interpersonal distance with teammates and game spatial references (targets). Possibly, these results may be linked to the effects of the differential-learning program, which seems to accelerate positional adjustments in younger teams who present a lower tactical knowledge. From a dynamic system point of view, differential learning induces more self-organized learning by increasing players' ability to use environmental information to sustain and coadapt immediate behaviors within teammates and against opponents (Travassos, Duarte, Vilar, Davids, & Araujo, 2012). That is, these challenging environments emphasize noise to promote the emergence of movement variability (Frank et al., 2008; Henz & Schöllhorn, 2016; Schöllhorn et al., 2009; Wagner & Muller, 2008), and therefore, players must be able to continuously adapt their behavior in space and time to the ever-changing environment by increasing the functionality of game decisions (Gonçalves et al., 2017b; Travassos et al., 2012). In this sense, it is possible that the incorporation of the differential-learning approach in dynamic contexts allows players to become more proficient at perceiving environmental cues and constant changes in game situations (Chow et al., 2015; Seifert et al., 2013). The general results indicated that while greater stability in players' distances is keeping with team structural shape (observed through trivial/small differences in absolute positional variables), the individual variability triggered by differential learning promoted completely different functional and auto-organized behaviors. Possibly, these results may be explained by recent works (Shirado & Christakis, 2017), which highlighted that local or individual variability may help to achieve global stability.

According to previous standpoints, the differential-learning effects were greater as far as the environment allows for capturing the qualitative changes of the degree of a movement pattern and promotes positional behavior adjustments. Overall, the differential-learning training program has distinct extents according to a team's age because it stressed creative and tactical behavior to a higher magnitude in the U13 players. In fact, dynamic

environmental contexts offer an alternative view to discover new potentials of the differential-learning approach. Nevertheless, a few limitations of this study are related to the absence of retention phase measurements, because several differential-learning studies have highlighted better performance scores in retention tests (Schöllhorn et al., 2006, 2012; Wagner & Muller, 2008). Further, the optimal stochastic perturbations pursuit is an essential requirement because the differential-learning training program induced distinct individual adaptations according to players' expertise levels. This result possibly indicated that different age groups should perform adapted fluctuations, and this requirement was not fulfilled in the current study because the exact same variations were applied for both U13 and U15 players. In addition, the differential learning embedded in SSG was only applied during the first 30 min, and the tasks performed during the remaining soccer training were not under control, which could have influenced some of the results. In forthcoming training programs, it will be important to understand the long-term impact of this approach on creative and tactical behavior. Hence, there is little guidance on how the learning process occurs throughout the training program—specifically if a longer training period results in greater advantages due to cumulative effects of prolonged practice or eventually leads to a stagnation phase. Thus, it could be important to add a few intermediate measurements between the pretest and posttest evaluations. Moreover, it will be interesting to incorporate a constant practice group to compare the effects with typical SSG and with the differential-learning approach embodied in noise game-based situations. Under these practice conditions, differential learning fosters skillfulness behaviors that play a key role in boosting a creative spark in youth soccer players.

Conclusions

In summary, this study demonstrated that a differential-learning training program embodied in SSG has great potential for fostering creative and tactical behavior in youth soccer players. Indeed, a training program sustained in differential learning facilitates the development of creativity components, mainly the originality, versatility, and attempts of movement actions. In addition, the differential-learning approach provided a substantial decrease in fails during game performance for both teams. Moreover, it seemed that the increase in individual variability promoted by the differential-learning approach favored the regularity of positional behavior in youth players compared with typical SSG formats. Accordingly, exposing players to dynamic

environmental contexts seems to emphasize their ability to identify and use environmental cues. An important finding from this study was that the results showed different extents according to age group, and therefore, differential learning may induce different adaptations based on a group's sport expertise. Considering the theoretical considerations of the differential-learning approach and the empirical results observed, a non-linear learning process highlighted important implications for the training process. Traditionally, differential learning has only been developed using isolated tasks, and this study showed that it could also improve players' performance using game-based situations. Therefore, coaches could use differential-learning variations sustained in SSG to foster players' creative and tactical behavior and to improve their ability to be attuned to the environment. Additionally, as creativity continues to be undeveloped, differential learning is a highly recommended approach that could be easily introduced into the physical education curriculum as it is in sports structures during the early ages.

What does this article add?

The current article provides preliminary scientific support for the importance of adaptive movement variability to facilitate the emergence of players' technical and tactical creativity. Exploration enriches variability of functional movement patterns and guides players to search for adaptive movement solutions to satisfy or overcome environmental fluctuations. However, teachers and coaches should be aware that these learning contexts require space and tolerance for players' errors and attempts, so they can nurture their creative predisposition. Hence, using stochastic perturbations in game-based situations appears necessary to facilitate adaptation to the different levels of complexity encountered in team sports.

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